Final Exam

● Graded

Student

Hangjia Gu

Total Points

27.5 / 45 pts

Question 1

Overall Grades **27.5** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 17.5 pts** Point adjustment

MATH 202-001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Hangjia Gu	hg182	Hangjia Gu

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- No Cellphones.
- No Calculators.
- Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: Hangjia Gu

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	2	5	3	1	5	1	0.5	5	27.5

Problem 1. [5'] Find the general solution of the linear system:

$$\begin{aligned} 2x - 3y &= 1 \\ -2y - 3z &= 1 \\ x + 2z &= 1 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right]$$

$R_3 \leftarrow R_3 - \frac{1}{2}R_1$

$$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & \frac{3}{2} & 2 & \frac{1}{2} \end{array}\right]$$

$R_3 \leftarrow R_3 + \frac{3}{4}R_2$

$$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & 0 & -1/4 & 7/4 \end{array}\right]$$

$R_1 \leftarrow R_1 - \frac{3}{2}R_2$

$$\left[\begin{array}{ccc|c} 2 & 0 & 9/4 & -1/2 \\ 0 & -2 & -3 & 1 \\ 0 & 0 & -1/4 & 7/4 \end{array}\right]$$

$R_1 \leftarrow R_1 + 9R_3$

$$\left[\begin{array}{ccc|c} 2 & 0 & 0 & 61/4 \\ 0 & -2 & -3 & 1 \\ 0 & 0 & -1/4 & 7/4 \end{array}\right]$$

$R_2 \leftarrow R_2 - 12R_3$

$$\left[\begin{array}{ccc|c} 2 & 0 & 0 & 61/4 \\ 0 & -2 & 0 & -20 \\ 0 & 0 & -1/4 & 7/4 \end{array}\right]$$

$R_1 \leftarrow \frac{1}{2}R_1$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 61/8 \\ 0 & -2 & 0 & -20 \\ 0 & 0 & -1/4 & 7/4 \end{array}\right]$$

$R_2 \leftarrow -\frac{1}{2}R_2$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 61/8 \\ 0 & 1 & 0 & 10 \\ 0 & 0 & -1/4 & 7/4 \end{array}\right]$$

$R_3 \leftarrow -4R_3$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 61/8 \\ 0 & 1 & 0 & 10 \\ 0 & 0 & 1 & -7 \end{array}\right]$$

$\Rightarrow$ general solution:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 61/8 \\ 10 \\ -7 \end{bmatrix}$$

General solution:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 11 \\ 7 \\ 5 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right]$$

~~$R_3 \leftarrow R_1 + 2R_3$~~

$R_3 \leftarrow -R_1 + 2R_3$

$$\sim \left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & 3 & 4 & 1 \end{array}\right]$$

$R_3 \leftarrow 3R_2 + 2R_3$

$$\sim \left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & 0 & -1 & 5 \end{array}\right]$$

$R_2 \leftarrow R_2 - 3R_3$

$$\sim \left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & 0 & -14 \\ 0 & 0 & -1 & 5 \end{array}\right]$$

$R_1 \leftarrow 2R_1 - 3R_2$

$$\sim \left[\begin{array}{ccc|c} 4 & 0 & 0 & 44 \\ 0 & -2 & 0 & -14 \\ 0 & 0 & -1 & 5 \end{array}\right]$$

$R_1 \leftarrow \frac{1}{4}R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & -2 & 0 & -14 \\ 0 & 0 & -1 & 5 \end{array}\right]$$

$R_2 \leftarrow -\frac{1}{2}R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & -1 & 5 \end{array}\right]$$

$R_3 \leftarrow -R_3$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -5 \end{array}\right]$$

$\Rightarrow x = 11,\ y = 7,\ z = -5.$

Check: $2x - 3y = 2 \times 11 - 3 \times 7 = 22 - 21 = 1$ ✓

$-2y - 3z = -2 \times 7 - 3 \times (-5) = -14 + 15 = 1$ ✓

$x + 2z = 11 + 2 \times (-5) = 11 - 10 = 1$ ✓

$\Rightarrow$ correct.

2/5

Problem 2. [5'] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

Set

$$\begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$$

$R_4 \leftarrow R_4 - \frac{1}{2}R_3$ why?

$$\sim \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix}$$

$R_1 \leftarrow R_1 + 2R_4$

$$\sim \begin{bmatrix} 4 & 0 & 0 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix}$$

As $R_2 = -5R_4$, R_2 is redundant.

$R_4 + \frac{1}{5}R_2$

$$\sim \begin{bmatrix} 4 & 0 & 0 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$R_1 \leftarrow \frac{1}{4}R_1$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$R_2 \leftarrow R_2 + 5R_1$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 5 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$R_2 \leftarrow \frac{1}{5}R_2$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$R_3 \leftarrow R_3 - R_2$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

general solution.

(1). Basis of column space of A:

pivot columns of A.

$$\Rightarrow v_1 = \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix} \quad v_2 = \begin{bmatrix} -5 \\ 0 \\ 5 \end{bmatrix} \quad v_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

v_1, v_2 and v_3 are linearly independent.

(2). Basis of row space of A:

nonzero rows in RREF

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

(3) Basis of null space A.

Solve: $Ax = 0$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Problem 3. [5] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

$A = QP^{-1}$

$\Rightarrow A =$ $\left[\begin{array}{ccc|ccc} 0 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 0 & 1 & 2 & 2 \\ -1 & -1 & -1 & & & \end{array}\right]$

$R_3 \leftarrow R_3 \times (-1)$

$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -1 & -1 \end{array}\right]$

$\Rightarrow P^{-1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$

$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 1 & 2 & 2 & 0 & 1 & 0 \end{array}\right]$

$A = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 2 & 2 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}^{-1}$

$\Rightarrow A = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$

$= \begin{bmatrix} -2 & -1 & -2 \\ 0 & -1 & -2 \end{bmatrix}$

Set P $\left[\begin{array}{ccc|ccc} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ -1 & -1 & -1 & 0 & 0 & 1 \end{array}\right]$

$\Rightarrow T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = \begin{bmatrix} -2 & -1 & -2 \\ 0 & -1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$\Rightarrow R_3 \leftarrow R_3 + R_2$

$\sim \left[\begin{array}{ccc|ccc} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 & 1 & 1 \end{array}\right]$

$R_3 \leftarrow R_3 + R_1$

$\sim \left[\begin{array}{ccc|ccc} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 1 & 1 \end{array}\right]$

~~$R_1 \leftarrow R_2$~~ $R_1 \leftarrow R_1 + R_2$

$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 1 & 1 \end{array}\right]$

$R_2 \leftarrow R_1 + R_2$

$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 & 2 & 0 \\ 0 & 0 & -1 & 0 & 1 & 1 \end{array}\right]$

row operation

$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & 1 \end{array}\right]$

3/5

Problem 4. [5] Find the Jordan canonical form of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

t

$$\begin{bmatrix} -\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ 2 & -2 & 1-\lambda \end{bmatrix} = 0$$

$$(2-\lambda)(-1)^{2+2}\begin{bmatrix} 1-\lambda & 0 \\ -2 & 1-\lambda \end{bmatrix} = 0$$

$$(2-\lambda)\left[(1-\lambda)^2 - 0\times(-2)\right] = 0$$

$$(2-\lambda)(1-\lambda)^2 = 0$$

$$(\lambda-2)(\lambda-1)^2 = 0$$

$\lambda_1 = 2$, with $AM_1 = 1$

$\lambda_2 = 1$. with $AM_2 = 2$

$$(A-\lambda I)v = 0$$

for λ_1,

$$\begin{bmatrix} -2 & 2 & 0 \\ 0 & 2-2 & 0 \\ 2 & -2 & 1-2 \end{bmatrix} \cdot v = 0$$

$$\begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ 2 & -2 & -1 \end{bmatrix} \cdot v = 0 \Rightarrow \begin{cases} -x+2y=0 \\ -2x-2y-z=0 \end{cases}$$

$$\Rightarrow \begin{cases} x=2y \\ z=-6y \end{cases}$$

for λ_2,

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = s\begin{bmatrix} 2 \\ 1 \\ -6 \end{bmatrix}, \ s\in\mathbb{R}. \quad v_1 = \begin{bmatrix} 2 \\ 1 \\ -6 \end{bmatrix}$$

$$GM(\lambda_1) = 1.$$

$$\begin{bmatrix} 1-1 & 2 & 0 \\ 0 & 2-1 & 0 \\ -2 & -2 & 1-1 \end{bmatrix} \cdot v = 0$$

$$\begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix} v = 0 \Rightarrow \begin{cases} 2y=0 \\ y=0 \\ -2x-2y=0 \end{cases} \Rightarrow \begin{cases} x=-y \\ y=0 \end{cases} \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ t \end{bmatrix}. \quad v_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

$GM(\lambda_2) = 2$.

~~$\Rightarrow$ GM = AM~~ $\Rightarrow GM(\lambda_1) = AM(\lambda_1) = 1$,

$GM(\lambda_2) = AM(\lambda_2) = 2$.

~~GM~~

$\Rightarrow$ all Jordan blocks are 1×1.

$\Rightarrow$ Jordan canonical form:

$$J(\lambda) = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}, \ J_2(\lambda) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

$$v_1(\lambda) = \begin{bmatrix} 2 \\ 1 \\ -6 \end{bmatrix} \quad v_2(\lambda) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow P = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 0 & 0 \\ -6 & 1 & 1 \end{bmatrix} \Rightarrow P^{-1} = ?$$

$$\left[\begin{array}{ccc|ccc} 2 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ -6 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & -1/2 & 1 & 0 \\ 0 & 1 & 1 & 3 & 0 & 0 \end{array}\right]$$

P is not invertable.

$A = PDP^{-1}$

Problem 5. [5'] Diagonalize the quadratic form

$$f(x, y, z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

$3x^2+3y^2+4z^2+2xz-2yz=0$

$\Rightarrow (x+z)^2+(y-z)^2+2(x^2+y^2+z^2)=0$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 1 & * \\ 0 & 1 & -1 & * \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$X^TAx = \sum_i a_{ii}x_i^2 + 2\sum_{i<j} a_{ij}x_ix_j$

$\Rightarrow f(x,y,z) = a_{11}x^2 + a_{12}y^2 + a_{33}z^2 + 2a_{12}xy + 2a_{13}xz + 2a_{23}yz$

$\Rightarrow a_{11}=3,\ a_{22}=3,\ a_{33}=4;$

$2a_{12}=0,\ 2a_{13}=2,\ 2a_{23}=-2;$

the rest $=0$

$$\Rightarrow A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -2 \\ 0 & 0 & 4 \end{bmatrix}.$$

$$\det\begin{bmatrix} 3-\lambda & 0 & 2 \\ 0 & 3-\lambda & -2 \\ 0 & 0 & 4-\lambda \end{bmatrix} = 0$$

$\Rightarrow (3-\lambda)(-1)^{2+2}\det\begin{bmatrix} 3-\lambda & 2 \\ 0 & 4-\lambda \end{bmatrix} = 0$

$(3-\lambda)\left[(3-\lambda)(4-\lambda) - 0\times 2\right] = 0$

$(3-\lambda)^2(4-\lambda) = 0$

$\lambda_1 = 3,\ AM(\lambda_1) = 2$

$\lambda_2 = 4,\ AM(\lambda_2) = 1.$

for v_1,

$$\det\begin{bmatrix} 3-3 & 0 & 2 \\ 0 & 3-3 & -2 \\ 0 & 0 & 4-3 \end{bmatrix} v_1 = \det\begin{bmatrix} 0 & 0 & 2 \\ 0 & 0 & -2 \\ 0 & 0 & 1 \end{bmatrix} v_1 = 0$$

$$\Rightarrow v_1 = t\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

for v_2,

$$\begin{bmatrix} 3-4 & 0 & 2 \\ 0 & 3-4 & -2 \\ 0 & 0 & 4-4 \end{bmatrix} v_2 = 0$$

$$\begin{bmatrix} -1 & 0 & 2 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix} v_2 = 0$$

$$v_2 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

$$\Rightarrow P = \begin{bmatrix} 1 & -2 & -2 \\ 1 & -2 & -2 \\ 0 & 1 & 1 \end{bmatrix}.$$

$$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

$PP^{-1} = I$

$$\left[\begin{array}{ccc|ccc} 1 & -2 & -2 & 1 & 0 & 0 \\ 1 & -2 & -2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

$R_2 \leftarrow R_2 - R_1$

$$\sim\left[\begin{array}{ccc|ccc} 1 & -2 & -2 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

$R_1 \leftarrow R_1 + 2R_2$

$$\sim\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

P is not invertable.

Problem 6. [5] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ such that

$$\begin{bmatrix} 0 & 2 & -1 \\ 2 & -1 & 1 \\ -2 & 2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

$$\Rightarrow \begin{cases} 2y - z = -3 \\ 2x - y + z = 8 \\ -2x + 2y = -8 \end{cases}$$

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right]$$

$R_1 \leftarrow R_1 - R_3$

$$\sim \left[\begin{array}{ccc|c} 2 & 0 & -1 & 5 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right]$$

$R_2 \leftarrow R_2 - R_1$

$$\sim \left[\begin{array}{ccc|c} 2 & 0 & -1 & 5 \\ 0 & -1 & 2 & 3 \\ -2 & 2 & 0 & -8 \end{array}\right]$$

$R_3 \leftarrow R_3 + 2R_2$

$$\sim \left[\begin{array}{ccc|c} 2 & 0 & -1 & 5 \\ 0 & -1 & 2 & 3 \\ -2 & 0 & 4 & -2 \end{array}\right]$$

$R_3 \leftarrow R_3 + R_1$

$$\sim \left[\begin{array}{ccc|c} 2 & 0 & -1 & 5 \\ 0 & -1 & 2 & 3 \\ 0 & 0 & 3 & 3 \end{array}\right]$$

$R_1 \leftarrow R_1 + \frac{1}{3}R_3$

$$\sim \left[\begin{array}{ccc|c} 2 & 0 & 0 & 6 \\ 0 & -1 & 2 & 3 \\ 0 & 0 & 3 & 3 \end{array}\right]$$

$R_2 \leftarrow R_2 - \frac{2}{3}R_3$

$$\sim \left[\begin{array}{ccc|c} 2 & 0 & 0 & 6 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 3 & 3 \end{array}\right]$$

$R_1 \leftarrow R_1 \times \frac{1}{2}$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 3 & 3 \end{array}\right]$$

$R_2 \leftarrow R_2 \times (-1)$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 3 & 3 \end{array}\right]$$

$R_3 \leftarrow R_3 \times \frac{1}{3}$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$$\Rightarrow \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}.$$

$\therefore a = 3,\ b = -1,\ c = 1.$

Check:

$$3\begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix} + (-1)\begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} + 1\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 6 \\ -6 \end{bmatrix} + \begin{bmatrix} -2 \\ 1 \\ -2 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$ result is correct.

Problem 7. [5'] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

solve:

$\vec{v} \cdot \vec{u} = 0$

$\|\vec{v}\| = 1$

$$\begin{bmatrix}0&3&0\\1&0&0\\1&2&0\\0&-1&0\end{bmatrix}$$

$$\begin{bmatrix}0&0&0\\1&0&0\\1&2&0\\0&-1&0\end{bmatrix}$$

Set $\vec{u} = [a, b, c, d]$

$$\begin{bmatrix}0&3\\1&0\\1&2\\0&-1\end{bmatrix}\begin{bmatrix}a&b&c&d\end{bmatrix} = 0$$

$\vec{v}$ is orthogonal to V

$\Rightarrow \vec{v} \cdot V = 0$ set $\vec{v} = [a\ b\ c\ d]$

$$\Rightarrow [a\ b\ c\ d]\begin{bmatrix}1&0\\1&2\\0&-1\end{bmatrix} = 0$$

$$\Rightarrow [b+c\ \ 2c-d] = 0$$

$$\Rightarrow \begin{cases} b = -c \\ 2c = d,\ d = -\frac{1}{2}c \end{cases}$$

$$\Rightarrow \begin{bmatrix}a\\b\\c\\d\end{bmatrix} = \begin{bmatrix}0\\-1\\1\\\frac{1}{2}\end{bmatrix} + t\begin{bmatrix}1\\0\\0\\0\end{bmatrix}$$

$$\Rightarrow \vec{v} = [0\ \ -1\ \ 1\ \ \tfrac{1}{2}]$$

$$q = \frac{u}{\|u\|} = \frac{[0\ -1\ 1\ \frac{1}{2}]}{\sqrt{0^2+(-1)^2+1^2+(\frac{1}{2})^2}}$$

$$= \frac{2}{3}\cdot[0\ -1\ 1\ \tfrac{1}{2}]$$

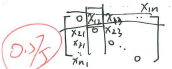
$$= [0\ -\tfrac{2}{3}\ \tfrac{2}{3}\ \tfrac{1}{3}]$$

$\Rightarrow$ an orthonormal basis for V is

$$\left\{\begin{bmatrix}0\\-\frac{2}{3}\\\frac{2}{3}\\\frac{1}{3}\end{bmatrix}, \begin{bmatrix}0\\-\frac{1}{3}\\\frac{1}{3}\\\frac{2}{3}\end{bmatrix}\right\}.$$

$$\Rightarrow \vec{v} = t\begin{bmatrix}1\\-5\\5\\\frac{1}{2}\end{bmatrix}$$

$$\vec{v_i} = v_i - \sum_{j=1}^{i-1} v_j u_i \cdot u_i$$



0.3/5

Problem 8. [5'] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

$A^T = -A \Rightarrow A$ is a skew-symmetric matrix.

which means $a_{nn} = 0$ for $n \in \mathbb{R}$.

$\det(A) = \sum_{j=1}^{n} a_{ij} \cdot C_{ij}$

When $i = j$, we have $a_{ii} \cdot C_{ii} = 0$ because $a_{ii} = 0$. ✓

The number of $a_{ij} \cdot C_{ij}$ that doesn't equal to 0 depends on the value of n.

When n is odd, then the 2 areas above and beneath a_{ii} has symmetry and cancel each other, giving $\det A = 0$.

Yes, but the expression is along a row, not symmetric ones.

Problem 9. [5'] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|$

$\Rightarrow (\vec{u}+\vec{v})^2 = (\vec{u}-\vec{v})^2$

$\|\vec{u}\|^2 + \|\vec{v}\|^2 + 2\vec{u}\cdot\vec{v} = \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2\vec{u}\cdot\vec{v}$

$\|\vec{u}\|^2 + \|\vec{v}\|^2 - (\|\vec{u}\|^2 + \|\vec{v}\|^2) + 2\vec{u}\cdot\vec{v} + 2\vec{u}\cdot\vec{v} = 0$

$4\vec{u}\cdot\vec{v} = 0.$

either $\|\vec{u}\| = 0$ or $\|\vec{v}\| = 0$,

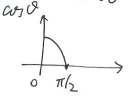
which means one of them is a zero vector.

zero vectors are always perpendicular to any other vector;

either we have $\|\vec{u}\| \neq 0$ and $\|\vec{v}\| \neq 0$,

but $\|\vec{u}\|\|\vec{v}\|\cdot\cos\theta = 0$

$\Rightarrow \cos\theta = 0$



$\Rightarrow \theta = \pi/2.$

$\Rightarrow \vec{u}$ and $\vec{v}$ are perpendicular to each other.